

GENERAL CONCEPTS

1. What Are Hormones?

Hormones are chemical messengers produced by specific tissues or cells. They are usually carried through the **blood (vascular system)** to other tissues, where they produce effects even at **very low concentrations**.

The endocrine system uses hormones to **coordinate, regulate, and control physiological processes** throughout the body.

Traditional definition of a hormone

A hormone is a chemical that:

- Is produced by a specific **endocrine organ or tissue**
- Is transported through the **blood**
- Acts on a **distant target organ or tissue**
- Produces an effect at **very low concentrations**

This definition is useful in veterinary medicine, but it is not completely strict because some substances considered hormones, such as **prostaglandins and somatomedins**, are produced by many different tissues rather than by one specific endocrine organ.

2. Other Chemical Signaling Systems

Not all chemical messengers travel through the blood. Some act mainly **locally**, allowing cells to communicate with nearby cells or with themselves.

A. Paracrine effectors

- The chemical messenger is released into the **interstitial fluid**.
- It usually acts on **nearby or adjacent cells**.
- If the messenger acts on the **same cell that produced it**, it is called an **autocrine effector**.

Remember:

- **Paracrine = acts on nearby cells**
- **Autocrine = acts on the cell that produced it**

B. Neurotransmitters

- These chemicals are used for communication:
 - between **neurons**, or
 - between **neurons and their target cells**
- Their action is generally limited to a **short distance** and a specific area of the target cell.

C. Exocrine effectors

- These substances are released into a **body cavity or duct**, rather than into the bloodstream.
- Example: substances produced by the **pancreas** that are released into the **gastrointestinal tract**.

3. Endocrine and Nervous Systems Work Together

The **endocrine system and nervous system are closely connected**.

- The **nervous system** is mainly responsible for **rapid control** of body functions.
- The **endocrine system** generally produces **slower but longer-lasting effects**.
- Together, they coordinate many physiological processes.

Example: Milk Let-Down Reflex

The milk let-down reflex shows how the nervous and endocrine systems work together.

Sequence:

1. Suckling occurs

→ The mammary gland is stimulated.

2. Nerve impulses are generated

→ Signals travel from the mammary gland to the **hypothalamus through the spinal tract**.

3. Hypothalamic neurons are stimulated

→ Neurosecretory neurons in the **supraoptic and paraventricular nuclei** synthesize **oxytocin**.

4. Oxytocin is transported

→ It travels down the axons of these neurons.

5. Oxytocin is released into the blood

→ It is released from nerve endings in the **posterior pituitary** into the bloodstream.

6. Oxytocin reaches the mammary gland

→ It causes contraction of **myoepithelial cells**.

7. Milk is pushed out

→ Myoepithelial cells surround the **alveoli**, which contain the milk-secreting cells.

→ Their contraction pushes milk into the **larger cisternae near the teat** and then into the **teat**.

Simple flow:

Suckling → Nerve impulses → Hypothalamus → Oxytocin release → Blood →

Mammary gland → Myoepithelial contraction → Milk let-down

4. Direct Interaction Between the Nervous and Endocrine Systems

The interaction between the two systems can be even more direct.

Example: Adrenal Medulla

- The **endocrine cells of the adrenal medulla** are directly controlled by **preganglionic neurons**.
- During **stress**, these neurons stimulate the adrenal medulla.
- The adrenal medulla then rapidly releases its hormones into the blood.

This allows the body to respond **immediately to stressful situations**.

Shared chemical transmitters

Some chemical messengers are found in **both nervous and endocrine tissues**, including:

- **Epinephrine**
- **Dopamine**
- **Histamine**
- **Somatostatin**

This further demonstrates the close relationship between the nervous and endocrine systems.

5. Major Functions Controlled by Hormones

The endocrine system helps regulate many important body functions, especially:

- **Metabolism**
- **Growth**
- **Reproduction**

A. Hormones involved in energy metabolism

These hormones help regulate how the body **uses and stores energy**:

- **Insulin**
- **Glucagon**
- **Cortisol**
- **Epinephrine**
- **Thyroid hormones**
- **Growth hormone**

B. Hormones involved in mineral metabolism

These help maintain proper **mineral and electrolyte balance**, especially calcium and sodium-related regulation:

- **Parathyroid hormone (PTH)**
- **Calcitonin**
- **Aldosterone**
- **Angiotensin**
- **Renin**

C. Hormones involved in growth

Growth is controlled by several hormones and growth factors, including:

- **Growth hormone**
- **Thyroid hormone**
- **Insulin**
- **Estrogen**
- **Androgens**
- Various **growth factors**

D. Hormones involved in reproduction

Important reproductive hormones include:

- **Estrogen**

- **Androgens**
- **Progesterone**
- **Luteinizing hormone (LH)**
- **Follicle-stimulating hormone (FSH)**
- **Prolactin (PRL)**
- **Oxytocin**

6. Signal Amplification in the Endocrine System

One important characteristic of hormones is **signal amplification**.

This means that a **very small amount of hormone can produce a large biological effect**.

Example with steroid hormones

A single steroid hormone molecule can activate a gene.

That can lead to:

1 hormone → many mRNA molecules → many enzyme molecules

Example with protein hormones

One protein hormone molecule can stimulate the production of many **cAMP molecules**.

Then:

1 hormone → many cAMP molecules → activation of many enzymes

Because of this amplification, hormones can produce strong effects even when present in extremely small amounts.

Important concentration

Hormones can produce significant effects at plasma concentrations as low as approximately:

10^{-11} to 10^{-12} mol

7. Hormones Can Affect Existing Enzyme Activity

Hormones do not only cause cells to make new proteins or enzymes.

They can also **change the activity of enzymes that are already present in the cell**.

This means that cells have some **basic or basal enzyme activity even when no hormone is present**. Hormones can increase or decrease this activity.

8. Speed and Duration of Hormonal Effects

Hormonal responses are generally:

- **Slower to begin**
- **Longer-lasting**

Hormonal effects may last from:

minutes → days

This is different from the nervous system.

Endocrine vs Nervous System

Feature	Endocrine System	Nervous System
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Main messenger	Hormones	Neurotransmitters/nerve signals
Transport	Usually through blood	Along neurons/synapses
Speed	Relatively slow	Very rapid
Duration	Longer-lasting	Usually short
Typical duration	Minutes to days	Milliseconds to seconds
Main role	Long-term or widespread regulation	Rapid control

Key idea to remember

Nervous system = fast and short-lasting

Endocrine system = slower and longer-lasting

However, the two systems are **not separate**. They work together to maintain and regulate normal physiological functions.

SYNTHESIS OF HORMONES

Hormones can be divided into several major classes:

Hormone class	Examples
Proteins	Growth hormone (GH), insulin, ACTH
Peptides	Oxytocin, vasopressin
Amines	Dopamine, melatonin, epinephrine
Steroids	Cortisol, progesterone, vitamin D

The way a hormone is produced and stored depends mainly on whether it is a **protein/peptide hormone** or a **steroid hormone**.

1. Protein and Peptide Hormone Synthesis

Basic process:

Preprohormone → Prohormone → Active hormone → Storage in granules → Exocytosis

Protein and peptide hormones are first made as **larger inactive precursor molecules** called **preprohormones**.

Step 1: Synthesis in ribosomes

- Hormone synthesis begins on **ribosomes**.
- The newly forming molecule is called a **preprohormone**.
- The **"pre" portion** directs the molecule to the **rough endoplasmic reticulum (RER)**.
- As the preprohormone is being made, it enters the **inside of the RER**.

Step 2: Formation of the prohormone

- Enzymes called **peptidases** in the RER quickly remove the **"pre" portion**.
- The remaining molecule is called a **prohormone**.

- The prohormone leaves the RER inside small **vesicles** that bud off from the RER.

Step 3: Processing in the Golgi apparatus

- The vesicles carrying the prohormone move to the **Golgi apparatus**.
- They fuse with the Golgi membranes and are processed into **secretory granules**.
- The prohormone is then **cleaved into its active hormone**.
- Most of the hormone reaches its final active form in the **Golgi apparatus**, although some prohormone may remain.

Step 4: Storage

- The active protein/peptide hormone is stored in **secretory granules inside the endocrine cell**.
- A small amount may be released continuously, but **most is stored until a specific signal tells the cell to release it**.

Step 5: Release by exocytosis

When the appropriate signal is received:

↑ Intracellular Ca^{2+} → granules fuse with cell membrane → exocytosis → hormone released

Exocytosis requires:

- **ATP** for energy
- **Calcium (Ca^{2+})**

The increase in cytoplasmic calcium can come from:

- Release of Ca^{2+} from the **endoplasmic reticulum**
- Release of Ca^{2+} from **mitochondria**
- **Influx of Ca^{2+} from outside the cell**

Easy flow to remember:

Ribosome → Preprohormone → RER removes "pre" → Prohormone → Golgi → Active hormone → Secretory granules → Ca^{2+} + ATP → Exocytosis → Blood

Key point:

Protein and peptide hormones are made first, stored in granules, and released when needed.

2. Steroid Hormone Synthesis

Steroid hormones are different from protein hormones.

Steroids are **lipophilic (fat-soluble)** and are mainly made from **cholesterol**.

Major steroid groups include:

Adrenocortical hormones

- **Glucocorticoids**
- **Mineralocorticoids**

Sex hormones

- **Estrogens**

- **Progesterone**
- **Androgens**

All steroid hormones have a common **four-ring, 17-carbon structure** that comes from cholesterol.

3. Where Does the Cholesterol Come From?

Most steroid-producing cells obtain cholesterol from the blood.

Main source:

Liver → cholesterol → blood → steroid-producing cell

The liver produces cholesterol, which can be transported in the blood as part of **low-density lipoproteins (LDLs)**.

How cholesterol enters the cell

1. **LDL binds to a receptor** on the cell membrane.
2. The LDL enters the cell.
3. **Lysosomal enzymes break down the LDL.**
4. The cholesterol is released.

The cholesterol can then either:

- Be used **immediately** to make steroids, or
- Be stored inside the cell as **cholesterol ester**.

A small amount of steroid precursor can also be synthesized within the cell from **acetate**, but most steroid hormones come from cholesterol.

4. How Steroid Hormones Are Made

The first step in making **all steroid hormones** from cholesterol is:

Cholesterol → Pregnenolone

This occurs in the **mitochondria**.

After pregnenolone is formed, it undergoes additional chemical changes. These reactions may occur:

- In the **mitochondria**, or
- In other parts of the cell after the steroid molecule moves there.

The exact control of steroid movement between different cell compartments is not completely understood.

5. Different Cells Make Different Steroid Hormones

The type of steroid hormone produced depends on the **enzymes present in the cell**.

In other words:

The enzymes determine which steroid hormone it can make.

Example: Adrenal cortex

Cells of the **adrenal cortex** have specific enzymes called **hydroxylases**.

These enzymes are needed to modify specific carbon positions, particularly:

- **Carbon 11**
- **Carbon 21**

These reactions are essential for producing:

- **Glucocorticoids**
 - **Mineralocorticoids**
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6. Sex Steroid Synthesis

Sex steroid production generally follows this sequence:

Pregnenolone → Progesterone → Androgens → Estrogens

However, different cells contain different enzymes, so they cannot all produce every type of steroid.

Example: Leydig cells of the testis

Leydig cells can:

- Make **pregnenolone**
- Make **progesterone**
- Convert progesterone into **androgens**

But they **lack the enzymes needed to convert androgens into estrogens**.

Therefore, Leydig cells mainly produce **androgens**, rather than estrogens.

7. Adrenal Cortex Can Also Make Sex Steroids

The adrenal cortex has enzyme systems that allow it to produce:

- **Adrenocortical hormones**
- **Sex steroids**

Normally, the adrenal cortex produces:

- **Large amounts of adrenocortical hormones**
- **Small amounts of sex steroids**

However, in certain **pathological conditions**, adrenal production of sex steroids can increase.

8. Steroid Hormones Are NOT Stored

This is one of the most important differences between steroid and protein hormones.

Protein/peptide hormones:

Stored in secretory granules → released when needed

Steroid hormones:

Not stored → synthesized and released immediately

Because steroids are **lipophilic**, they can easily pass through the cell membrane by **simple diffusion**.

Therefore:

Rate of steroid hormone secretion = rate of steroid hormone synthesis

The two processes are therefore **tightly coupled**.

What is stored?

Steroid-producing cells do not store large amounts of the finished hormone.

The main stored material is the **precursor cholesterol**, stored as **cholesterol ester**.

PROTEIN/PEPTIDE VS STEROID HORMONES

Feature	Protein/Peptide Hormones	Steroid Hormones
Starting material	Amino acid chains/protein precursors	Cholesterol
Initial synthesis	Ribosomes	Mainly mitochondria + other cell compartments
Precursor	Preprohormone → prohormone	Cholesterol → pregnenolone → other steroids
Storage	Stored in secretory granules	Not stored as finished hormone
Release	Exocytosis	Simple diffusion
Main requirement for release	Ca²⁺ and ATP	No exocytosis required
Secretion	Can be stored and released when signaled	Usually released as soon as synthesized
Main control of secretion	Release of stored hormone	Rate of synthesis

HIGH-YIELD SUMMARY

Protein and peptide hormones

Ribosome → Preprohormone → RER → Prohormone → Golgi → Active hormone → Granules → Ca²⁺ + ATP → Exocytosis

- Made as **preprohormones**
- RER removes the "pre" portion
- Prohormone is processed in the **Golgi**
- Stored in **secretory granules**
- Released by **exocytosis**
- Release requires **Ca²⁺ and ATP**

Steroid hormones

Cholesterol → Pregnenolone → Steroid hormone → Diffusion out of cell

- Mainly derived from **cholesterol**
- First step occurs in the **mitochondria**
- Specific **enzymes determine which steroid is produced**
- **Not stored as finished hormones**
- Released by **simple diffusion**
- **Synthesis and secretion occur together**
- Cholesterol can be stored as **cholesterol ester**

TRANSPORT OF HORMONES IN THE BLOOD

Hormones that act on distant target tissues must be transported through the **bloodstream**.

The way a hormone is carried in the blood depends mainly on whether it is **water-soluble (hydrophilic)** or **fat-soluble (lipophilic)**.

1. Protein Hormones Are Hydrophilic

Protein and peptide hormones are hydrophilic, meaning they **dissolve easily in water**.

Since blood plasma is mostly water, these hormones can travel through the bloodstream in **dissolved form**.

Forms of protein hormones in the blood

Protein hormones may circulate as:

- **Monomers** = one unit
- **Polymers** = multiple units

For example, **insulin** can occur in different forms.

Some hormones that normally have several subunits may also circulate as **individual subunits**.

However:

Subunit form → lower biological activity/potency

This means that separating the subunits generally makes the hormone **less effective**.

Key point:

Protein/peptide hormones = hydrophilic → dissolve directly in plasma → usually do not need carrier proteins.

2. Steroid and Thyroid Hormones Are Lipophilic

Steroid hormones and thyroid hormones are lipophilic, meaning they are **fat-soluble**.

Because blood is an aqueous (water-based) environment, these hormones have **poor solubility in plasma**.

Therefore, they usually cannot travel freely through the blood in large amounts.

Instead, they are transported by **binding to plasma proteins**.

3. Types of Transport Proteins

There are two important types of protein binding involved in transporting steroid and thyroid hormones:

A. Specific/high-affinity binding proteins

These proteins have a **strong preference for particular hormones**.

Example:

Transcortin

- Has a high affinity for **cortisol**
- Also has a high affinity for **corticosterone**

- It can also transport **progesterone**, although it binds progesterone less strongly

Important characteristic:

High affinity = strong binding but low capacity

These proteins have **low capacity** because they are present in relatively **small amounts in plasma**.

B. Nonspecific/low-affinity binding proteins

The main example is **albumin**.

- Albumin binds steroid hormones with **low affinity**.
- However, albumin is present in **very high concentrations in plasma**.

Therefore:

Low affinity = weaker binding but high capacity

Albumin can carry **large amounts of steroid hormones** because there is so much albumin in the blood.

4. Free Hormone vs Bound Hormone

A hormone can exist in the blood in two forms:

Bound hormone

Hormone attached to a transport protein.

Free (unbound) hormone

Hormone that is not attached to a protein.

The **free form is the biologically active form**.

A hormone must become **free/unbound** before it can enter a target cell and produce its biological effect.

Important relationship:

Bound hormone $\rightleftharpoons$ Free hormone

There is a continuous **equilibrium** between the two forms.

When free hormone is used by tissues, more bound hormone quickly **dissociates from its carrier protein** and becomes free.

This allows the body to maintain a supply of active hormone.

5. How Much Hormone Is Free?

Only a **small portion** of a lipophilic hormone in the blood is usually free.

General rule:

About 1% of the total hormone is free

The remaining majority is **bound to proteins**.

Exception: Cortisol

Cortisol has a somewhat larger free fraction:

Up to about 10% of cortisol may be free.

Remember:

Most lipophilic hormone = bound

Small amount = free and biologically active

6. Why Protein Binding Is Important

Binding to carrier proteins provides several advantages:

- It allows **lipophilic hormones to be transported through watery blood**.
- It creates a **reservoir of hormone** that can quickly release free hormone.
- It helps maintain a relatively stable concentration of free hormone.

The free hormone is continually used by tissues, and the bound hormone can **dissociate quickly** to replace it.

7. Measurement of Hormone Levels

For most hormones, laboratories usually measure the **total hormone concentration**, meaning:

Free hormone + Bound hormone = Total hormone

Exception: Thyroid hormones

For **thyroid hormones**, there is often greater interest in determining the amounts of:

- **Bound hormone**
- **Free hormone**

because the relationship between these forms is especially important when evaluating thyroid function.

8. Steroid Hormone Concentration Reflects Secretion

For steroid hormones:

Synthesis → Release → Blood concentration

Steroid hormone synthesis and secretion are **tightly linked** because steroids are generally **not stored in large amounts** inside endocrine cells.

Also, their **metabolic clearance rates are usually relatively constant**.

Therefore:

Plasma steroid concentration can often give a good indication of the rate at which the steroid is being secreted.

9. Physiological Conditions Can Change Hormone Transport

The relationship between hormone concentration and secretion can change when the body changes the amount of **hormone-binding proteins**.

Example: Pregnancy in humans

During pregnancy:

- Production of **estrogen increases**
- Estrogen can increase the production of **estrogen-binding proteins**
- More hormone may therefore become bound
- This can affect the **metabolism and clearance of estrogen**

Thus, physiological conditions can change hormone levels not only by changing hormone production but also by changing **binding proteins and hormone metabolism**.

HIGH-YIELD COMPARISON

Feature	Protein/Peptide Hormones	Steroid & Thyroid Hormones
Solubility	Hydrophilic	Lipophilic
Blood transport	Dissolved directly in plasma	Bound to plasma proteins
Need carrier proteins?	Generally no	Yes
Free hormone	Circulates mainly dissolved	Usually only a small fraction
Active form	Free hormone	Free hormone
Main binding proteins	Usually not important for transport	Specific proteins + albumin
Example	Insulin	Cortisol, progesterone, thyroid hormones

TYPES OF BINDING PROTEINS

High-affinity proteins

Strong binding + low capacity

Example:

Transcortin → cortisol

Albumin

Weak binding + high capacity

Reason:

Albumin is present in **very large amounts in plasma**.

EASY FLOW TO REMEMBER

Protein/peptide hormones:

Hydrophilic → dissolve in plasma → circulate freely

Steroid/thyroid hormones:

Lipophilic → cannot dissolve well in plasma → bind carrier proteins → small free fraction → free hormone enters target cells → biological effect

Most important concept:

BOUND = transport/reservoir

FREE = biologically active

HORMONE-CELL INTERACTION

A hormone only affects certain cells even though it may circulate throughout the body. This is because **target cells have specific receptors** that recognize and bind the hormone.

1. Location of Hormone Receptors

The location of the receptor depends on the type of hormone.

Protein and peptide hormones

- These hormones **cannot pass through the cell membrane** because they are water-soluble.
- Their receptors are located on the **plasma membrane** of the target cell.

Steroid hormones

- Steroids are **lipid-soluble**, so they can pass through the cell membrane.
- Their receptors are found inside the cell, in the:
 - **Cytoplasm**, or
 - **Nucleus**

Easy rule:

Protein/peptide → receptor on the cell membrane

Steroid → receptor inside the cell

2. Why Only Certain Cells Respond to Hormones

Although some hormones, especially steroids, can enter many or even most cells, they only produce an effect in cells that have the **correct receptor**.

Hormone receptors have two important characteristics:

Specificity

A receptor recognizes and binds a **particular hormone**.

High affinity

The receptor can bind the hormone **strongly**, even when the hormone is present in very small amounts in the blood.

Therefore:

Specific receptor + high affinity → small hormone concentration can still produce a strong biological response

3. Termination of Hormone Action

A hormone's effect must eventually stop.

The most common way this happens is when the hormone **dissociates (separates) from its receptor**.

Hormone-receptor binding is **noncovalent**, meaning the hormone and receptor are not permanently chemically attached.

Usually:

↓ Hormone concentration in blood → more hormone leaves receptor → hormone action decreases/stops

When the hormone concentration falls, the chemical equilibrium favors **dissociation** rather than continued binding.

4. Receptor Internalization

Hormone action can also stop when the **hormone-receptor complex is taken into the cell by endocytosis**.

Inside the cell:

- **Lysosomal enzymes** break down the hormone.
- The receptor is protected because it is associated with the membrane of the vesicle.
- The receptor can then be **recycled back to the plasma membrane** and used again.

Simple sequence:

Hormone + receptor → endocytosis → hormone degraded → receptor recycled

5. Number of Receptors

Cells normally have **many more receptors than are actually needed** to produce a biological response.

In fact:

Less than 50% of receptors usually need to be occupied to produce a maximal response.

This means a cell can still respond maximally even when many receptors are unoccupied.

6. Receptor Regulation

Although cells have excess receptors, the **number of receptors can change**.

Changes in receptor number mainly affect the cell's **sensitivity** to a hormone, rather than its maximum possible response.

More receptors

↑ Receptors → ↑ chance of hormone-receptor interaction → ↑ sensitivity

Fewer receptors

↓ Receptors → ↓ chance of interaction → ↓ sensitivity

7. Upregulation: Increasing Receptor Numbers

A hormone can sometimes stimulate production of **more receptors for a different hormone**.

Example: Ovarian follicle

During the later part of the follicular phase:

- **FSH** stimulates ovarian granulosa cells.
- FSH causes the cells to increase their **LH receptors**.
- The predominant receptor changes from **FSH receptors to LH receptors**.

This allows control of the ovarian follicle to shift from **FSH to LH**.

This change is important for:

- **Ovulation**
- **Luteinization**, which is the formation of the **corpus luteum**

Key concept:

FSH → increases LH receptors → follicle becomes more responsive to LH → ovulation + luteinization

8. Downregulation: Decreasing Receptor Numbers

Continuous exposure to a hormone can cause the number of its receptors to **decrease**.

This is called **downregulation**.

It may occur when:

- A hormone or agonist continuously stimulates the receptor.
- An agonist has a **very high affinity** for the receptor.
- Hormone levels become **pathologically high**.

Result:

Persistent high hormone exposure → ↓ receptor numbers → ↓ cell sensitivity → hormone resistance

This can cause an animal to become **less responsive or resistant to continued hormone treatment**.

POSTRECEPTOR CELL RESPONSES

After a hormone binds to its receptor, the cell must produce a response.

The mechanism depends on whether the hormone is a **steroid** or a **protein/peptide hormone**.

9. Steroid Hormone Action

Steroid hormones can cross the plasma membrane because they are **lipid-soluble**.

They can then interact with receptors in the **cytoplasm or nucleus**.

A. Steroids with cytoplasmic receptors

Examples:

- **Androgens**
- **Glucocorticoids**

The sequence is:

Steroid enters cell → binds cytoplasmic receptor → receptor becomes activated → complex moves into nucleus → binds DNA → mRNA produced → protein synthesis → biological response

Step-by-step:

1. The steroid passes through the **plasma membrane**.
 2. It binds its specific receptor in the **cytoplasm**.
 3. The hormone-receptor complex becomes **activated**.
 4. The complex moves into the **nucleus**.
 5. It binds specific regions of DNA called **response elements** on the chromatin.
 6. This stimulates the production of **messenger RNA (mRNA)**.
 7. mRNA moves to the **ribosomes**.
 8. Ribosomes use the mRNA to make specific **proteins/enzymes**.
 9. These proteins produce the desired **biological response**.
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B. Steroids with nuclear receptors

Some steroids, such as **estrogen**, can enter the nucleus and bind directly to their **nuclear receptors**.

The basic outcome is still:

Steroid → receptor → DNA response element → mRNA → protein synthesis → biological effect

10. Why Steroid Effects Are Usually Slower

Steroid hormones generally produce effects by changing **gene expression and making new proteins**.

Because protein synthesis takes time:

Steroid hormone action = relatively slow but often longer-lasting

11. Protein and Peptide Hormone Action

Protein and peptide hormones **cannot cross the plasma membrane**.

Therefore, they cannot directly enter the cell and interact with intracellular DNA receptors like steroids do.

Instead, they bind to **receptors on the plasma membrane**.

The receptor then activates an internal signaling molecule called a **second messenger**.

Basic pathway:

Hormone → membrane receptor → second messenger → intracellular enzymes/proteins → biological response

12. cAMP: An Important Second Messenger

One of the best-known second messengers is **cyclic AMP (cAMP)**.

Sequence:

Hormone binds receptor → activates adenylyl cyclase → cAMP increases → protein kinases activated → cellular response

Step-by-step:

1. Protein/peptide hormone binds to its **receptor on the plasma membrane**.
2. This activates the enzyme **adenylyl cyclase**.
3. Adenylyl cyclase produces **cAMP**.
4. cAMP activates **protein kinases**.
5. Protein kinases **phosphorylate proteins**.
6. These changes produce the **biological response**.

13. Other Second Messengers

cAMP is not the only second messenger.

Important examples include:

Calcium (Ca^{2+})

- Changes in cytoplasmic calcium can act as an important intracellular signal.

- Calcium can work with proteins such as **calmodulin**.

IP_3 (inositol triphosphate)

- Produced from **phosphatidylinositol metabolism**.
- One major action of IP_3 is to stimulate the release of **Ca^{2+} from intracellular stores**.

DAG (diacylglycerol)

- Also produced from **phosphatidylinositol metabolism**.
- Can activate **phospholipase A**.
- This promotes formation of **arachidonic acid**.
- Arachidonic acid is then used to produce **prostaglandins**.

Important flow:

$\text{IP}_3 \rightarrow \uparrow \text{intracellular } \text{Ca}^{2+}$

DAG → phospholipase A activation → arachidonic acid → prostaglandins

14. Protein/Peptide Hormones Usually Act Faster

Protein and peptide hormones often produce a **faster response than steroid hormones**.

Why?

Because they can activate **enzymes and proteins that already exist inside the cell**.

They usually do **not need to wait for the cell to make new enzymes**.

Compare:

Protein/peptide hormone:

Hormone → membrane receptor → second messenger → activates existing enzymes → **rapid response**

Steroid hormone:

Hormone → intracellular receptor → DNA → mRNA → new protein synthesis → **slower response**

HIGH-YIELD COMPARISON

Feature	Protein/Peptide Hormones	Steroid Hormones
Can cross cell membrane?	No	Yes
Receptor location	Plasma membrane	Cytoplasm or nucleus
Main mechanism	Second messengers	Gene regulation
Examples	Insulin, peptide hormones	Cortisol, estrogen, androgens
Major signaling molecules	cAMP, Ca^{2+} , IP_3 , DAG	Hormone-receptor complex
Main cellular effect	Activates existing proteins/enzymes	Stimulates new protein synthesis
Speed	Usually faster	Usually slower

HIGH-YIELD RECEPTOR CONCEPTS

Specificity

Correct receptor = correct hormone response

High affinity

Strong receptor-hormone binding → effective even at low hormone concentrations

Upregulation

↑ receptor number → ↑ sensitivity

Example:

FSH → ↑ LH receptors in granulosa cells

Downregulation

↓ receptor number → ↓ sensitivity/resistance

Example:

Persistently high hormone/agonist exposure → receptor loss → reduced response

Most important distinction

STEROIDS:

Enter cell → bind intracellular receptor → affect DNA → make new proteins

PROTEIN/PEPTIDE HORMONES:

Bind membrane receptor → use second messenger → activate existing cell machinery

METABOLISM OF HORMONES

Hormone activity is limited by how quickly hormones are **metabolized and removed from the body**.

The metabolism process is different for **steroid, thyroid, and protein hormones**.

1. Steroid Hormone Metabolism

Steroid hormones are mainly metabolized in the **liver**.

The usual process is:

Steroid → reduction → conjugation → increased water solubility → excretion in urine

Step 1: Reduction

The steroid molecule is chemically modified, usually by **reduction**.

Step 2: Conjugation

The modified steroid is combined with:

- **Sulfates**
- **Glucuronides**

This process is called **conjugation**.

Why is conjugation important?

Steroid hormones are naturally **lipid-soluble**. Conjugation makes them much more **water-soluble**.

Therefore:

Water-soluble steroid metabolites → easier removal → excreted in urine

Main organ:

Liver

2. Thyroid Hormone Metabolism

During metabolism of **thyroid hormones**, **iodine molecules are removed** from the hormone.

This process is called **deiodination**.

So:

Thyroid hormone → removal of iodine → metabolites

3. Protein Hormone Metabolism

Protein hormones are broken down by enzymes called **peptidases**.

If the protein hormone contains **disulfide bonds**, these bonds are first reduced before the protein is cleaved.

Basic sequence:

Disulfide bond reduction → peptidase cleavage → smaller hormone fragments/metabolites

4. Are Hormone Metabolites Always Inactive?

Usually, a hormone metabolite is **less biologically active** than the original hormone. However, this is **not always true**.

Some steroid metabolites or conjugates may still have **significant biological activity**.

Example 1: Testosterone → Dihydrotestosterone (DHT)

Testosterone → DHT

This conversion occurs inside certain tissues.

The question is whether this should be considered "metabolism" because:

- DHT is produced from testosterone.
- However, **DHT is more biologically potent than testosterone**.

Therefore, not every metabolic conversion makes a hormone less active.

Example 2: Estradiol → Estrone

Peripheral tissues, including **adipose tissue**, can convert:

Estradiol-17β → Estrone

This is considered a form of hormone metabolism.

However:

- **Estrone is a natural estrogen**
- It is still **relatively potent biologically**

Key point:

Metabolism does not always mean complete loss of biological activity.

5. Hormone Clearance

Clearance refers to how quickly a hormone is removed from the blood.

In most situations, hormone metabolism and clearance are relatively **constant**.

Therefore, the concentration of a hormone in the blood usually reflects its **rate of synthesis/secretion**.

Simple relationship:

↑ **Hormone synthesis/secretion** → ↑ **Blood hormone concentration**

↓ **Hormone synthesis/secretion** → ↓ **Blood hormone concentration**

This is because the rate at which the hormone is normally cleared tends to remain fairly stable.

6. Factors That Can Change Hormone Clearance

Although clearance is usually relatively constant, it can change under certain conditions.

Pregnancy

In humans, pregnancy can increase the production of **hormone-binding plasma proteins**.

More hormone may become bound to these proteins, which can **decrease hormone clearance**.

Liver disease

Liver disease can decrease the amount of hormone-binding proteins.

This may **increase hormone clearance** in some situations.

Important idea:

Changes in **binding proteins** can alter the rate at which hormones are cleared from the circulation.

FEEDBACK CONTROL MECHANISMS

Hormone concentrations must be carefully controlled because **the effects of hormones depend on their concentration in the blood**.

The main factor determining blood hormone concentration is generally the **rate of hormone secretion**.

The body uses **feedback systems** to keep hormone levels within an appropriate range.

7. Negative Feedback

Negative feedback is the most important and most common hormone control system.

In negative feedback:

↑ **Hormone concentration** → ↓ **further hormone production/secretion**

The purpose is to prevent hormone levels from becoming too high and to help maintain a **stable internal environment**.

Basic principle:

Too little hormone → **increase production**

Enough/too much hormone → **decrease production**

This creates a continuous adjustment toward the **optimal hormone concentration**.

8. Hypothalamus-Pituitary Negative Feedback

The **hypothalamus** and **anterior pituitary gland** are important control centers for many endocrine systems.

When hormone levels are too low:

↓ **Target gland hormone**

→ hypothalamus detects the low level

→ ↑ **releasing hormone**

→ anterior pituitary releases more **tropic hormone**

→ target endocrine organ is stimulated

→ ↑ **target hormone production**

When hormone levels are too high:

↑ **Target gland hormone**

→ hypothalamus decreases **releasing hormone**

→ anterior pituitary decreases **tropic hormone**

→ target organ decreases hormone production

→ ↓ **target hormone**

General flow:

Low target hormone → ↑ **Hypothalamic releasing hormone** → ↑ **Pituitary tropic hormone** → ↑ **Target gland hormone**

High target hormone → ↓ **Hypothalamic releasing hormone** → ↓ **Pituitary tropic hormone** → ↓ **Target gland hormone**

9. Negative Feedback Is Continuous, Not "All-or-None"

The endocrine system does not simply switch hormone production completely **on or off**.

Instead, hormone production is **continuously adjusted**.

The system constantly monitors hormone concentrations and makes small changes to keep the level near the desired **physiological set point**.

Key concept:

Negative feedback = continuous adjustment toward an optimal level

10. Negative Feedback Can Also Control Physiological Substances

Negative feedback is not limited to situations where one hormone controls another hormone.

A physiological substance can also trigger a hormonal response that brings the substance back toward normal.

Example: Blood glucose and insulin

↑ **Blood glucose** → ↑ **insulin secretion** → **glucose is reduced toward normal**

Although increased glucose causes **more insulin**, this is still considered negative feedback because the **final effect of insulin is to decrease the original increase in glucose**.

Simple idea:

Change in physiological variable → hormone response → variable moves back toward normal

11. Positive Feedback

Positive feedback is much less common than negative feedback.

In positive feedback:

Initial stimulus → increased hormone response → stronger stimulus → even greater hormone release

Instead of opposing the original change, the system **amplifies it**.

12. Example of Positive Feedback: LH Surge Before Ovulation

A classic example is the **preovulatory LH surge**.

Sequence:

Ovarian follicle develops

→ follicle produces increasing amounts of **estrogen**

→ increased estrogen stimulates the hypothalamic-pituitary system

→ **LH secretion increases greatly**

→ large **LH surge**

→ **ovulation**

→ the stimulus for the LH surge decreases

Thus:

↑ Estrogen → ↑ LH → ovulation → estrogen stimulus declines

The LH surge is therefore a **positive-feedback response** to increasing estrogen.

Important point:

The LH response does not continue forever. There is a definite endpoint:

Ovulation

The duration of the LH surge is also influenced by control mechanisms within the **hypothalamus**.

NEGATIVE VS POSITIVE FEEDBACK

Feature	Negative Feedback	Positive Feedback
Main effect	Opposes the original change	Amplifies the original change
Purpose	Maintains stability	Produces a major, usually temporary event
Frequency	Very common	Less common
Example	↑ Glucose → ↑ insulin → ↓ glucose	↑ Estrogen → ↑ LH → ovulation
End result	Returns variable toward normal	Drives process toward a specific endpoint

ENDOCRINE SECRETORY PATTERNS

Hormones are not always released at a constant rate.

Their secretion can follow **rhythmic patterns** that are influenced by factors such as:

- **Sleep**
- **Light**
- Other environmental or physiological signals

13. Circadian Rhythm

A **circadian rhythm** is a biological rhythm that follows an approximately **24-hour cycle**.

Many daily hormone rhythms are influenced by **light or darkness**.

Important terminology:

Circadian rhythm

= approximately 24-hour rhythm and is the preferred general term.

Diurnal rhythm

= specifically refers to a rhythm associated with activity during the **daytime**.

Nocturnal rhythm

= refers to a rhythm that is active mainly at **night**.

Therefore, not every 24-hour rhythm should automatically be called "diurnal."

14. Ultradian Rhythm

Hormonal changes can also occur at much **shorter intervals**.

A rhythm occurring repeatedly within a day, often approximately **every hour**, is called an:

Ultradian rhythm

Example concept:

Instead of one large hormone release each day, a hormone may be released in **repeated pulses throughout the day**.

HIGH-YIELD SUMMARY

Hormone Metabolism

Steroids

Reduction → sulfate/glucuronide conjugation → ↑ water solubility → urinary excretion

- Mainly metabolized by the **liver**
- Conjugation makes steroids easier to eliminate

Thyroid hormones

Iodine is removed during metabolism

Protein hormones

Disulfide bonds may be reduced → peptidases cleave the hormone

Important exception:

Metabolites are **not always inactive**.

Examples:

- **Testosterone** → **DHT**, which is more potent
- **Estradiol** → **estrone**, which remains a natural and relatively potent estrogen

FEEDBACK CONTROL

Negative feedback

Most common

↑ **Hormone** → ↓ **further hormone secretion**

Main purpose:

Maintain stable hormone levels

Positive feedback

Less common

↑ **Stimulus** → ↑ **hormone** → **stronger response**

Example:

↑ **Estrogen** → **LH surge** → **ovulation**

SECRETORY RHYTHMS

Circadian = approximately **24-hour rhythm**

Diurnal = related to **daytime**

Nocturnal = related to **nighttime**

Ultradian = shorter rhythms, often occurring **about every hour**

One-Line Exam Reminders

Steroid metabolism:

Reduction + conjugation → **water-soluble** → **urine**

Protein hormone metabolism:

Peptidases break them down

Negative feedback:

High hormone → **decrease production**

Positive feedback:

Stimulus increases response

Circadian:

~24 hours

Ultradian:

Shorter than 24 hours